

Increasing Logarithmic Weights for Rolling Validation: Simulation Study

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Purpose

The companion theory report, *Increasing Logarithmic Weights for Rolling Validation: Theory* (08A), derives asymptotic expressions for the expectation, variance, and Chebyshev ratio of several increasing logarithmic weighting schemes. This report examines whether those patterns are visible at finite sample sizes and whether the empirical wrong-selection probability decreases accordingly.

The notation and sign convention match 08A. The statistic is denoted by Ξ_n ; a positive value favors the oracle, and $\Xi_n < 0$ is incorrect selection. The weights appear throughout in the same order:

- 1.
2. $\log(i + 1)$;
3. $[\log(i + 1)]^2$;
4. $1 + \log(i + 1) + [\log(i + 1)]^2$;
5. $\log \log(i + e)$.

Simulation design

The simulation design and cached results are inherited from the authoritative 07 analysis, with the decreasing weights removed from this report. There are 5,000 independent replications, seed 2026, processed in batches of 250. The sample sizes are 100, 200, 500, 1000, 2000, 5000, 10000, 20000, 50000, 100000. Each Gaussian path is reused across weights and sample sizes, so comparisons are dependent. Exact moment trajectories extend from $n = 50$ through 10^7 on an approximately log-spaced grid that includes the simulation sizes.

The original implementation checks the quadratic-form representation, exact moments, direct score calculations, and the variance recurrence. The largest absolute discrepancy across these checks is $5.7\text{e-}14$. No Monte Carlo experiment is rerun here; 08B reads the validated 07 cache and filters it to the five weights above.

Table 1: Simulation design inherited from the authoritative 07 analysis.

Item	Value
Replications	5,000
Seed	2026
Batch size	250
Simulation sample sizes	100 to 100,000 (10 specified sizes)
Exact-trajectory range	50 to 10,000,000
Path reuse	Across weights and sample sizes

Do empirical moments agree with theory?

The first check compares exact moments with Monte Carlo moments. Agreement supports both the theoretical formulas used in 08A and their numerical implementation. Separate vertical scales are used because the five weights produce moments of very different magnitudes.

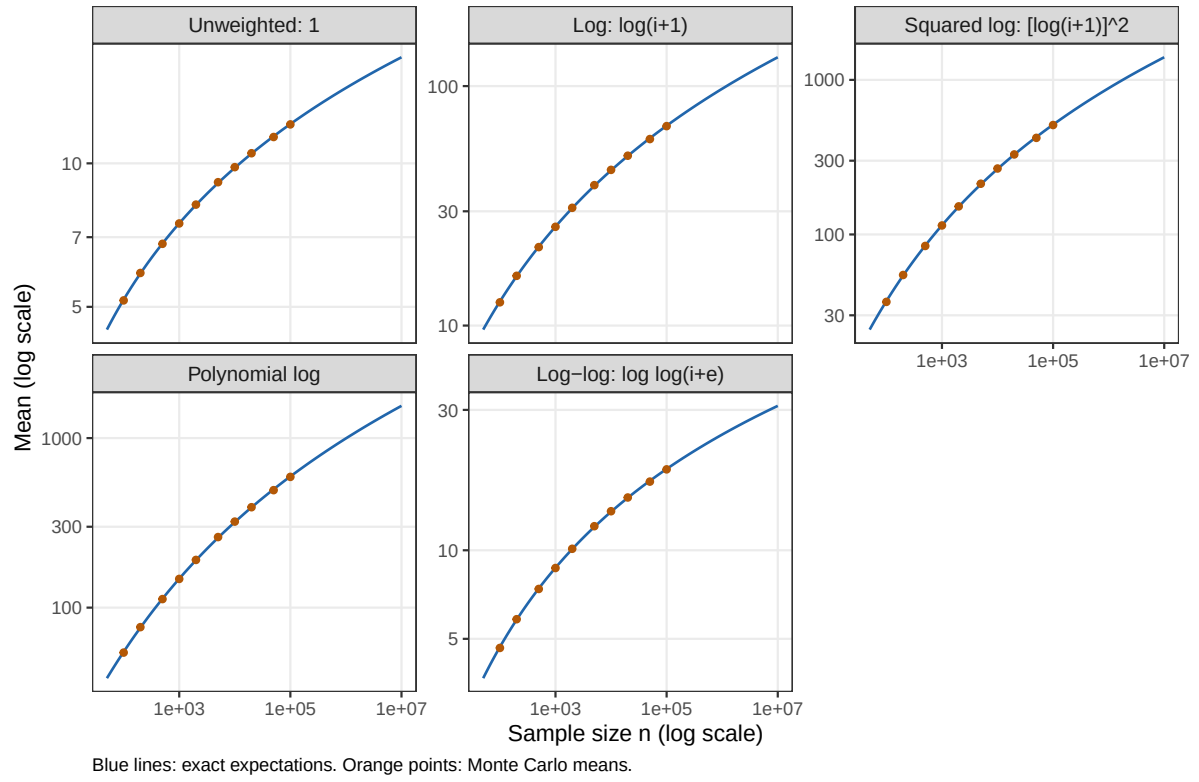


Figure 1: Exact and empirical expectations for the five weights.

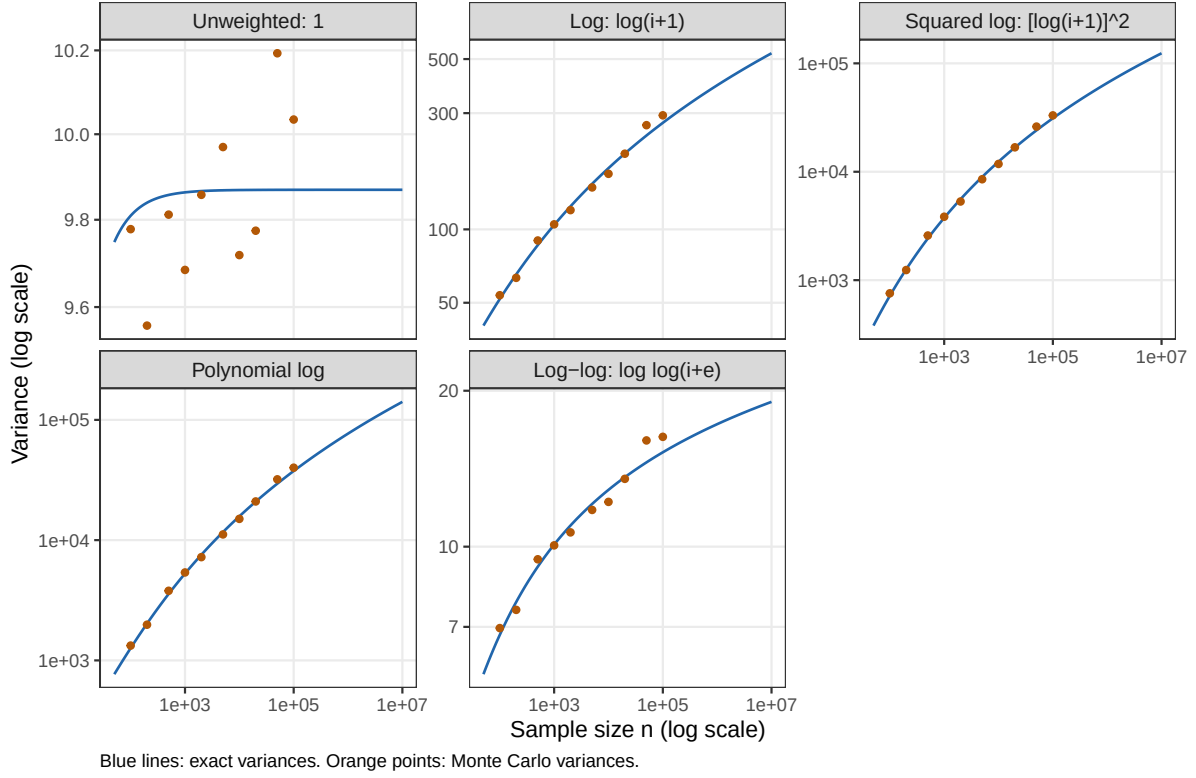


Figure 2: Exact and empirical variances for the five weights.

Across the displayed sample sizes, the empirical means closely follow the exact expectations. Empirical variances fluctuate more because they estimate a second moment, but they track the exact variance trajectories. The polynomial-log moments grow on the same leading scale as the squared-log moments, as predicted by the highest-degree-term result in 08A.

Does the Chebyshev ratio follow the prediction?

The theory in 08A predicts a vanishing Chebyshev ratio for all five weights. The blue curves below are the exact ratios. The orange points replace the exact variance by the Monte Carlo variance while retaining the exact mean in the denominator.

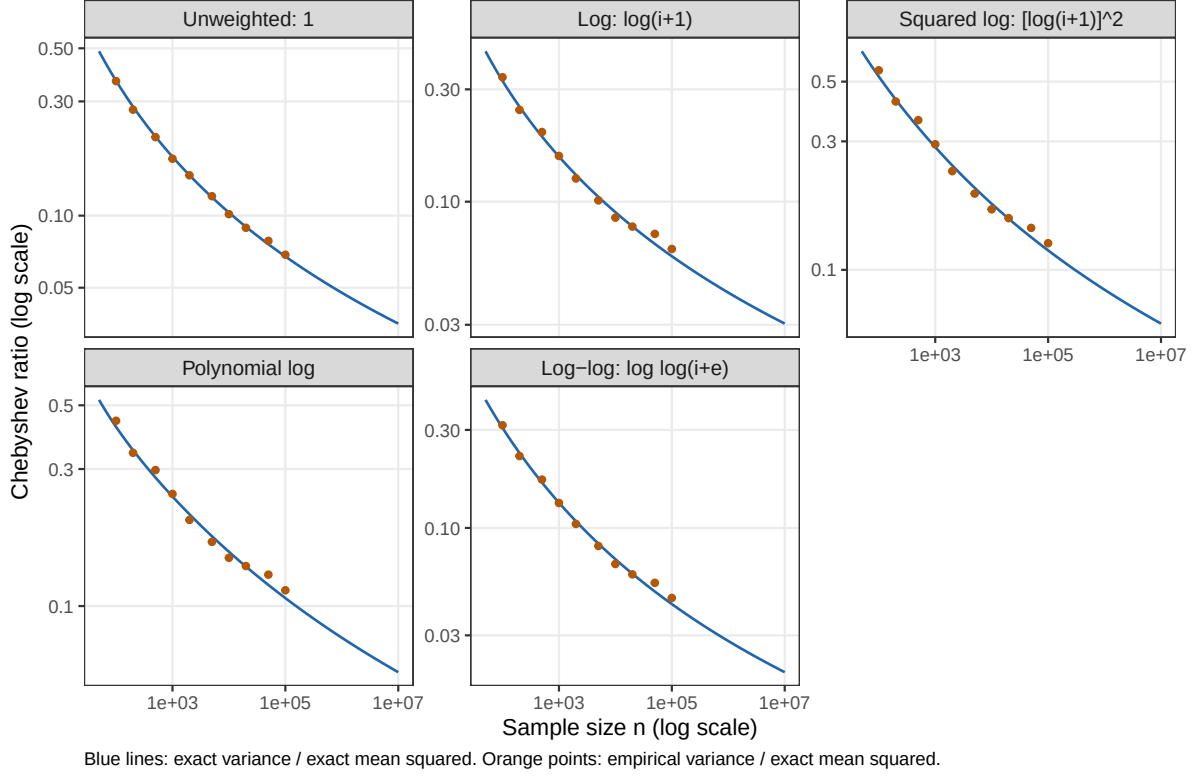


Figure 3: Exact and empirical Chebyshev ratios.

Every exact ratio decreases over the displayed range. The empirical points follow the same broad pattern. The squared-log and polynomial-log curves are similar but not identical at finite n ; their agreement is asymptotic because the polynomial contains lower-degree terms. The log-log ratio also decreases, despite the exceptionally slow growth of its weights.

Does the wrong-selection frequency decrease?

The wrong-selection frequency estimates the actual finite-sample probability $\mathbb{P}(\Xi_n < 0)$. It is not the Chebyshev ratio. The ratio supplies a sufficient upper-bound argument for asymptotic consistency, whereas the frequency below directly records observed selection mistakes. Bars are pointwise 95% Wilson intervals.

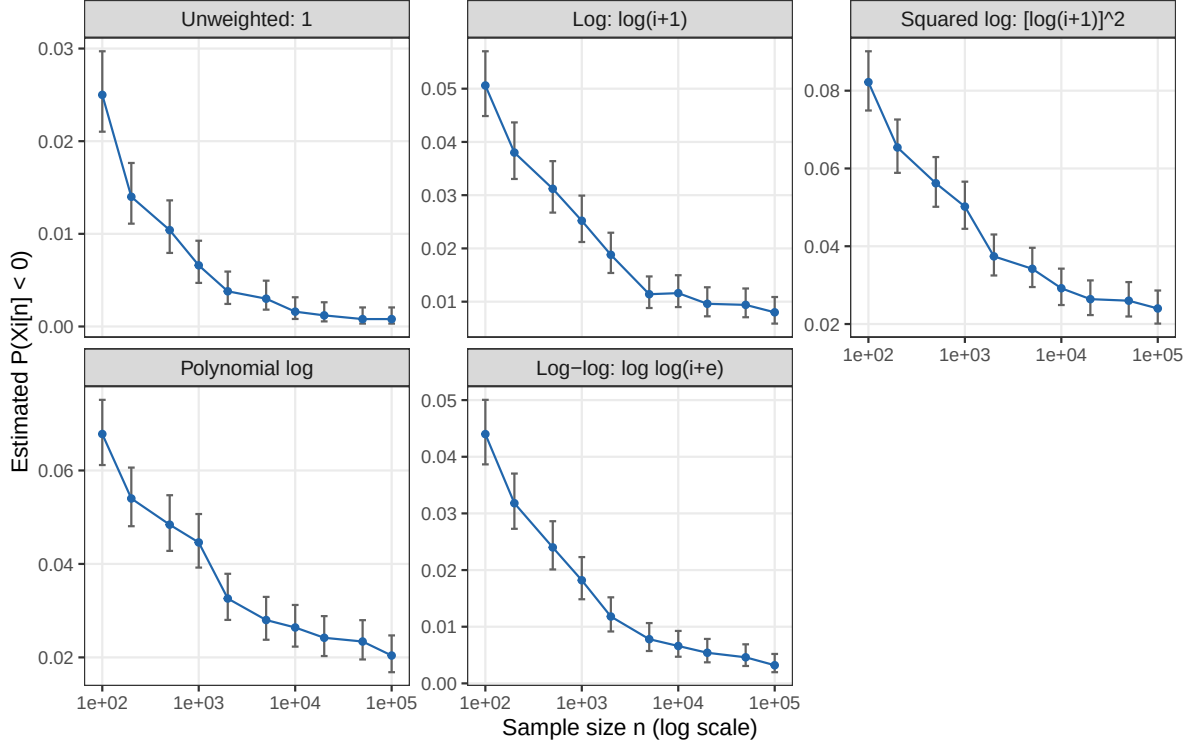


Figure 4: Wrong-selection frequencies and pointwise Monte Carlo uncertainty.

Wrong-selection frequencies decline for every weight. The decline is slowest for the squared-log and polynomial-log choices over the simulated range, which is compatible with their larger finite-sample variability. The log and log-log weights also show decreasing error frequencies, while the unweighted case serves as a low-error reference.

Table 2: Exact moments, Monte Carlo moments, ratios, and wrong-selection frequencies at $n = 100,000$.

Weight	Exact mean	MC mean	Exact variance	MC variance	Exact ratio	Wrong-selection frequency	Wilson lower	Wilson upper
Unweighted: 1	12.0901	12.0707	9.8695	10.0347	0.0675	0.0008	0.0003	0.0021
Log: $\log(i+1)$	68.3192	68.0434	273.7620	293.8490	0.0587	0.0080	0.0059	0.0109
Squared log: $[\log(i+1)]^2$	513.4996	510.3664	31143.6081	33119.3697	0.1181	0.0240	0.0201	0.0286
Polynomial log	593.9089	590.4804	37596.7760	40033.3312	0.1066	0.0204	0.0168	0.0247
Log-log: $\log \log(i+e)$	18.8973	18.8386	15.2116	16.2936	0.0426	0.0032	0.0020	0.0052

Weight-specific interpretation

Log weight. The exact mean and variance grow on the scales predicted in 08A, and both the Chebyshev ratio and wrong-selection frequency decline.

Squared-log weight. The empirical moments follow the exact trajectories. The ratio and error frequency decrease more slowly at the available sample sizes than for the log weight, but the direction agrees with the theory.

Polynomial-log weight. Its trajectories resemble those of squared log, while retaining visible finite-sample differences from the lower-degree terms. This is the expected approach to highest-degree-term dominance.

Log-log weight. Both the ratio and wrong-selection frequency decline even though the weight grows extremely slowly, matching the qualitative conclusion of 08A.

Summary

The simulations support the theoretical moment calculations for all five weights. Exact and empirical Chebyshev ratios decline, and the observed wrong-selection frequencies decrease across the simulated sample sizes. The squared-log and polynomial-log cases show slower finite-sample improvement, while the log-log case already exhibits the predicted downward behavior.

These numerical results do not replace the asymptotic proofs in 08A. They show how the theoretical mechanisms appear at finite sample sizes and preserve the distinction between a sufficient Chebyshev argument and the directly observed wrong-selection frequency.